# Master of Science in Data Science Study Plan

## Program Requirements

1. Compulsory courses (15 credit hours)
2. Elective Courses (9 credit hours)
3. Thesis (9 credit hours)

## Study Plan: Course List

| Course Code | Course Title | اسم المساق | Credit Hours | Pre-requisite |
|---|---|---|---|---|
| | **Compulsory Courses** | | | |
| 1440581 | Regression Modelling | نماذج الانحدار | 3 | 1440281 or equiv. |
| 1440582 | Introduction to Bayesian Data Analysis | مقدمة في تحليل بييز للبيانات | 3 | 1440281 or equiv. |
| 1501565 | Data Mining | التنقيب عن البيانات | 3 | (1501263 or 1501567 or equiv.) |
| 1501564 | Foundations of Data Science | اساسيات علم البيانات | 3 | (1511263 or 1501567 or equiv.) And 1440281. |
| 1501591 | Research Methodology | طرائق البحث | 3 | (1501215 or 1501501) and Graduate Standing |
| 1501696 | Thesis in Data Science | أطروحة علم البيانات | 9 | 1501591 |
| **Elective Courses** | | | | |
| **Mathematics Electives** | | | | |
| 1440583 | Graphical data analysis | تحليل البيانات الرسومية | 3 | 1440281 or equiv. |
| 1440584 | Applied Time Series Analysis | المتسلسلات الزمنية التطبيقية | 3 | 1440281 or equiv. |
| 1440586 | Design of Experiments | تصميم التجارب | 3 | 1440281 or equiv. |
| 1440589 | Nonparametric Inference of Statistics | الاستدلال اللامعلمي للإحصاء | 3 | 1440281 or equiv. |
| 1440590 | Stochastic Processes | العمليات العشوائية | 3 | 1440281 or equiv. |
| 1440593 | Topics in Statistics | موضوعات في الإحصاء | 3 | 1440281 or equiv. |
| **Data-centric Electives** | | | | |
| 1501668 | Big Data and Data Analytics | البيانات الضخمة وتحليلات البيانات | 3 | 1501263 or 1501567 or equiv. |
| 1501661 | Topics in Data Analytics and Cloud Computing | موضوعات في تحليلات البيانات و الحوسبة السحابية | 3 | Graduate Standing |
| 1501664 | Topics in Data Science | موضوعات في علم البيانات | 3 | 1501263 or 1501567 or equiv. |
| **Machine Intelligence Electives** | | | | |
| 1501530 | Advanced Artificial Intelligence | موضوعات في الذكاء الاصطناعي | 3 | Graduate standing |
| 1501531 | Machine Learning | التعلم الآلي | 3 | 1440211 + 1501215 |
| 1501511 | Advanced Programming | البرمجة المتقدمة | 3 | Basic programming course |
| 1501663 | Information Retrieval | استرجاع المعلومات | 3 | 1501263 or 1501567 or equiv. |

# Study Plan: Course Distribution

**First Year**

**Fall Semester**

| Course # | Course Title | Cr.Hr |
|---|---|---|
| 1501565 | Data Mining | 3 |
| 1440581 | Regression Modelling | 3 |
| | Elective Course | 3 |
| | **Total** | **9** |

**Spring Semester**

| Course # | Course Title | Cr.Hr |
|---|---|---|
| 1501564 | Foundations of Data Science | 3 |
| 1501591 | Research Methodology | 3 |
| 1440582 | Introduction to Bayesian Data Analysis | 3 |
| | **Total** | **9** |

**Second Year**

**Fall Semester**

| Course # | Course Title | Cr.Hr |
|---|---|---|
| | Elective Course | 3 |
| | Elective Course | 3 |
| 1501696 | Thesis in DS | 3 |
| | **Total** | **9** |

**Spring Semester**

| Course # | Course Title | Cr.Hr |
|---|---|---|
| 1501696 | Thesis in DS | 6 |
| | | |
| | | |
| | **Total** | **6** |